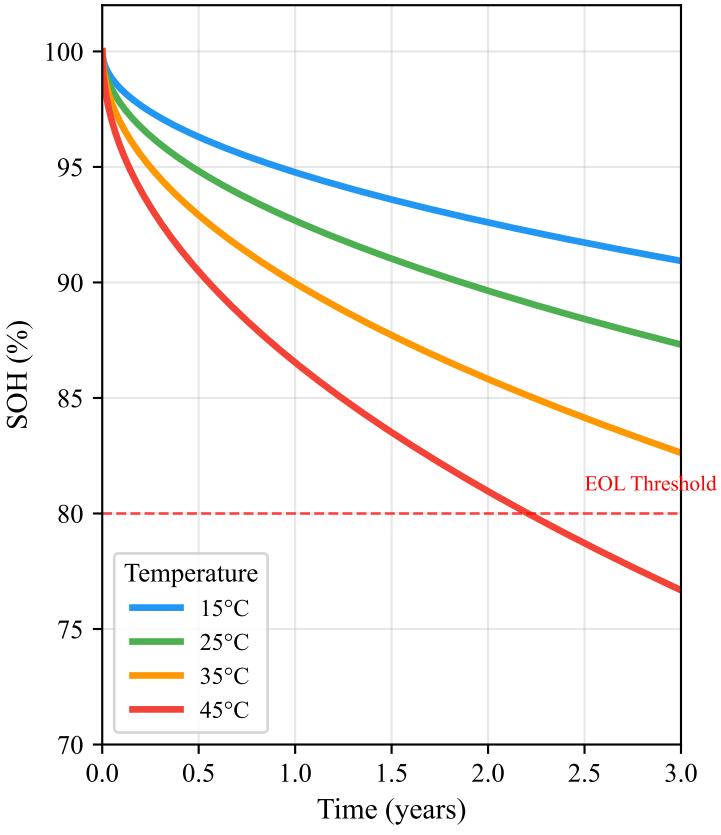

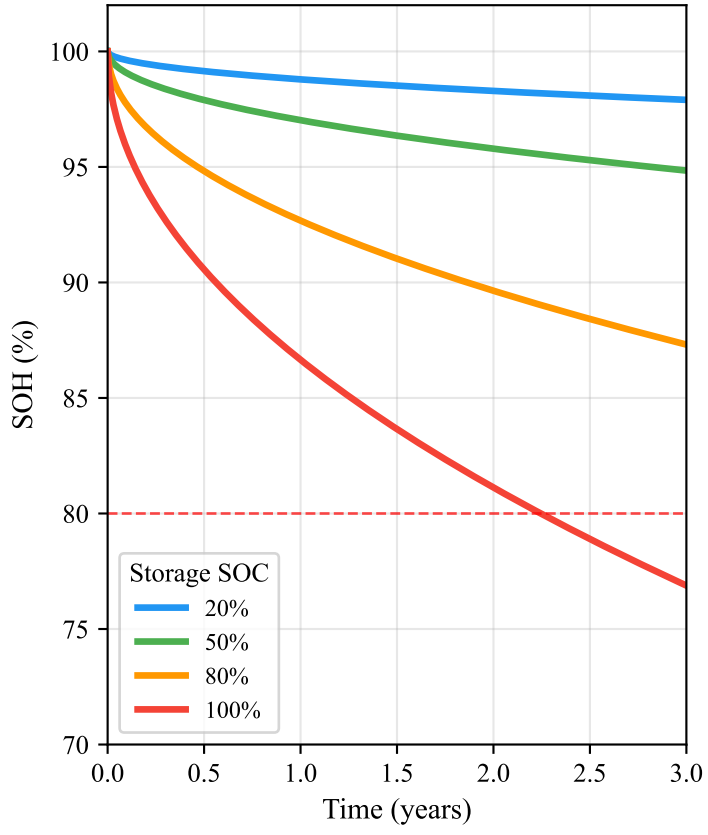

Calendar Aging Model

$$\frac{dSOH_{cal}}{dt} = -k_{cal}(T, SOC)$$

$$k_{cal} = A_{cal} \cdot \exp\left(-\frac{E_a}{R \cdot T}\right) \cdot f_{SOC}$$

$$f_{SOC} = \exp(\beta \cdot (SOC - 0.5))$$

Parameters :

$A_{cal} = 8.5 \times 10^{-2}$ $s^{-0.5}$

$E_a = 24$ kJ/mol

$\beta = 3.0$ (SOC stress factor)

Key Insights :

- High T accelerates SEI growth
- High SOC increases side reactions
- $\sqrt{t}$ dependence (diffusion-limited)